

Project : Healthcare Analytics for Improving Patient Outcomes

Overview

This project focuses on analysing hospital operations and patient data to uncover trends, evaluate resource utilisation, and provide actionable insights for improving service quality and patient outcomes. By leveraging data analytics techniques, the goal is to optimise hospital processes, improve patient care, enhance operational efficiency, and identify areas for resource optimisation. The analysis will include descriptive statistics, trend analysis, and visualisation to aid decision-making.

Problem Statement

The objective of this project is to analyse hospital and patient data to identify trends and patterns that influence key outcomes, including patient length of stay, billing amounts, resource allocation, and service quality.

The project will focus on the following areas:

- Assessing how patient demographics and medical conditions correlate with resource utilisation.
- Analysing billing patterns and identifying cost-saving opportunities.
- Evaluating hospital operations by looking at variables like admission type, discharge dates, and room utilisation.
- Providing insights for hospital management to improve operational efficiency and patient care.

Deliverables

1. Exploratory Data Analysis (EDA) : Conduct an in-depth EDA to uncover key trends, identify outliers, and visualise the relationships between hospital variables and patient outcomes.
2. Data Preprocessing : Clean the data by handling missing values, transforming categorical variables, and standardising formats to ensure consistency in analysis.
3. Trend Analysis : Use descriptive statistics and visualisation techniques to analyse trends related to patient length of stay, admission type, billing amounts, and resource utilisation.
4. Key Metrics : Analyse and report on important metrics, such as:

- Average patient billing amounts
- Room utilisation rates
- Length of stay analysis
- Admission type distribution (urgent vs. non-urgent)

5. Recommendations : Provide actionable recommendations based on the data analysis to improve hospital efficiency, reduce costs, and enhance patient care.

6. Final Report : A comprehensive report documenting the methodology, insights, and actionable recommendations, including visualisations to support conclusions.